

I. Yerga kiruvchi radar (Ground-Penetrating Radar)

Yerga kiruvchi radar (GPR) yer ostidagi ob'ektlarni aniqlash va ularning joylashuvini topish uchun ishlatiladi. Buning uchun elektromagnit to'lqinlar yerga yuboriladi va shu ob'ektlardan qaytgan to'lqinlar qabul qilinadi. Antenna va detektor to'g'ridan-to'g'ri yer ustida joylashgan bo'lib, ular bir nuqtada bo'ladi.

Burchak chastotasi ω bo'lgan chiziqli qutblangan elektromagnit tekis to'lqin z yo'nalishida tarqaladi va uning maydoni quyidagi ifoda bilan ifodalanadi:

$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z),$$

bu yerda E_0 — o'zgarmas, α — so'nish koeffitsienti (attenuation coefficient), β — to'lqin soni (wave number) bo'lib, ular mos ravishda quyidagicha ifodalanadi:

$$\alpha = \omega \left\{ \frac{\mu\epsilon}{2} \left[\left(1 + \frac{\sigma^2}{\epsilon^2 \omega^2} \right)^{1/2} - 1 \right] \right\}^{1/2}, \quad \beta = \omega \left\{ \frac{\mu\epsilon}{2} \left[\left(1 + \frac{\sigma^2}{\epsilon^2 \omega^2} \right)^{1/2} + 1 \right] \right\}^{1/2}$$

bu yerda μ — magnit singdiruvchanlik (magnetic permeability), ϵ — dielektrik o'tkazuvchanlik (electrical permittivity), σ — elektr o'tkazuvchanlik (electrical conductivity).

Radar signalining amplitudasi boshlang'ich qiymatining $1/e$ ($\approx 37\%$) dan pastga tushganda signal aniqlanmay qoladi. Odatda o'zgaruvchan chastotali (10 MHz – 1000 MHz) elektromagnit to'lqin ishlatiladi, bu aniqlash masofasi va rezolyutsiyasini (resolution) sozlash imkonini beradi.

GPR samaradorligi uning rezolyutsiyasiga bog'liq. Rezolyutsiya — aniqlanishi kerak bo'lgan ikkita qo'shni qatlam (reflector) orasidagi minimal masofa. Minimal masofa detektorda ikki qaytgan to'lqin orasidagi minimal faza farqi 180° ni ta'minlashi kerak.

Savollar: (Berilgan: $\mu_0 = 4\pi \times 10^{-7}$ H/m va $\epsilon_0 = 8.85 \times 10^{-12}$ F/m)

1. Yer magnit xususiyatga ega emas ($\mu = \mu_0$) va

$$\left(\frac{\sigma}{\omega\epsilon} \right)^2 \ll 1$$

sharti bajariladi deb faraz qiling. (1) va (2) tenglamalardan foydalanib, tarqalish tezligi ν ning μ va ϵ orqali ifodasini keltirib chiqaring. [1.0 ball]

2. Yer o'tkazuvchanligi $\sigma = 1.0$ mS/m va dielektrik o'tkazuvchanligi $\epsilon = 9\epsilon_0$ bo'lib,

$$\left(\frac{\sigma}{\omega\epsilon} \right)^2 \ll 1$$

sharti bajariladi deb faraz qiling ($\mu = \mu_0$). Ob'ektni aniqlashning maksimal chuqurligini aniqlang. [2.0 ball]

3. Yerga gorizontol holda ko'milgan ikkita parallel o'tkazgich sterjenni qaraylik. Sterjenlar 4 metr chuqurlikda joylashgan. Yerning o'tkazuvchanligi $\sigma = 1.0$ mS/m va dielektrik o'tkazuvchanligi $\epsilon = 9\epsilon_0$ ga teng. Faraz qilaylik, GPR o'lchovi taxminan sterjenlardan birining ustida amalga oshirilmoqda. Nuqtaviy detektor ishlatiladi deb hisoblang. 50 sm lateral rezolyutsiyaga (yonma-yon ajratish) erishish uchun zarur bo'lgan minimal chastotani aniqlang. [3.5 ball]

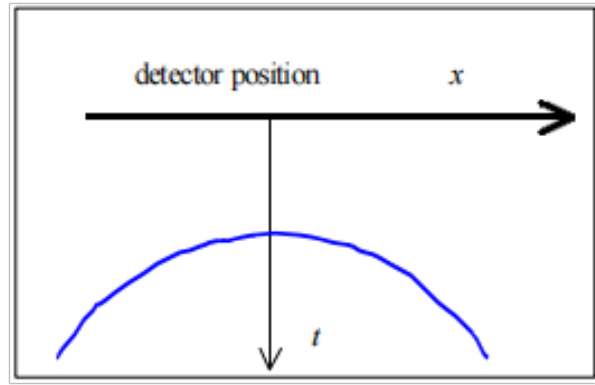


Figure 1: Grafik: t vaqtning x masofaga bog‘liqligi

4. Yerdagi bir sterjenning chuqurligi d ni aniqlash maqsadida, sterjenga perpendikulyar chiziq bo‘ylab o‘lchovlar o‘tkaziladi. Quyidagi rasmda t (ikki yo‘l vaqti, travel time) ning detektor holati x ga bog‘liqlik grafigi keltirilgan. $t_{\min} = 100$ ns.

t ni x funksiyasi sifatida keltirib chiqaring va d ni aniqlang. [3.5 ball]

II. Elektr signallarini sezish (Sensing Electrical Signals)

Ba’zi dengiz hayvonlari boshqa jonzotlarni ma’lum masofadan aniqlay olish qobiliyatiga ega. Buning sababi – nafas olish yoki mushak qisqarishi bilan bog‘liq jarayonlarda hosil bo‘ladigan elektr toklaridir. Ba’zi yirtqichlar bu elektr signallardan foydalanib, o‘ljalarini, hatto ular qum ostida ko‘milgan bo‘lsa ham, joylashtirishadi.

O‘lja tok hosil bo‘lishi va yirtqich tomonidan uni aniqlashning fizik mexanizmini quyidagicha modellashtirish mumkin (II-1-rasm). O‘lja tanasidagi musbat va manfiy potensialga ega bo‘lgan ikkita shar orqali o‘lja tomonidan hosil qilingan tok oqadi. Sharlar markazlari orasidagi masofa l_s , har bir sharning radiusi r_s bo‘lib, $r_s \ll l_s$. Dengiz suvining qarshiligi (resistivity) ρ . O‘lja tanasining qarshiligi atrofdagi dengiz suvi qarshiligiga teng deb faraz qilinadi, shuning uchun rasmdagi o‘ljani o‘rab turgan chegara (boundary) e’tiborsiz qoldirilishi mumkin.

O‘lja tomonidan hosil qilingan elektr quvvatini yirtqich tomonidan aniqlashni tavsiflash uchun detektor ham xuddi shunday ikki shar bilan modellanadi. Bu sharlar yirtqich tanasida joylashgan bo‘lib, atrofdagi dengiz suvi bilan kontaktida va o‘lja tanasidagi juftlikka parallel holda joylashgan. Ular bir-biridan l_d masofada ajratilgan, har birining radiusi r_d bo‘lib, $r_d \ll l_d$. Bunda detektor markazi manba ustida y masofada joylashgan va ikki sharni birlashtiruvchi chiziq elektr maydoniga parallel, II-1-rasmda ko‘rsatilganidek. l_s va l_d ham y dan ancha kichik. Ikki sharni birlashtiruvchi chiziq bo‘ylab elektr maydon kuchi o‘zgarmas deb hisoblanadi. Shuning uchun detektor II-2-rasmda tasvirlanganidek, o‘lja, atrofdagi dengiz suvi va yirtqichni birlashtiruvchi yopiq zanjir tizimini hosil qiladi.

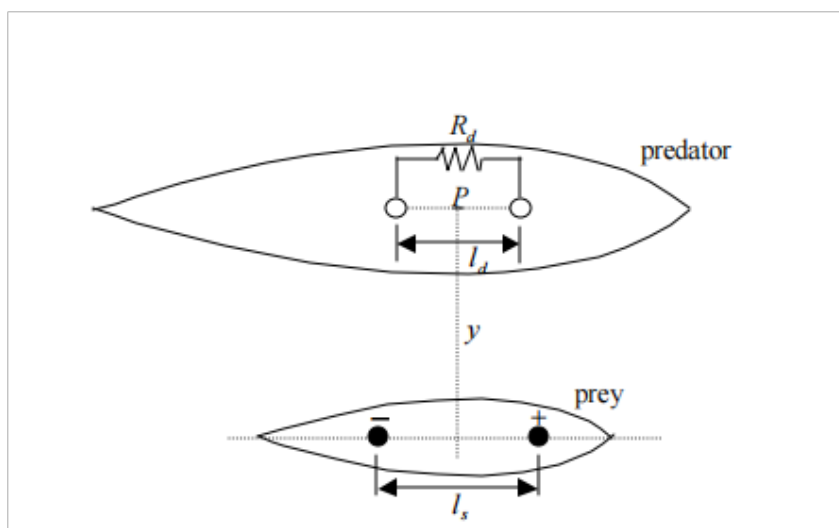
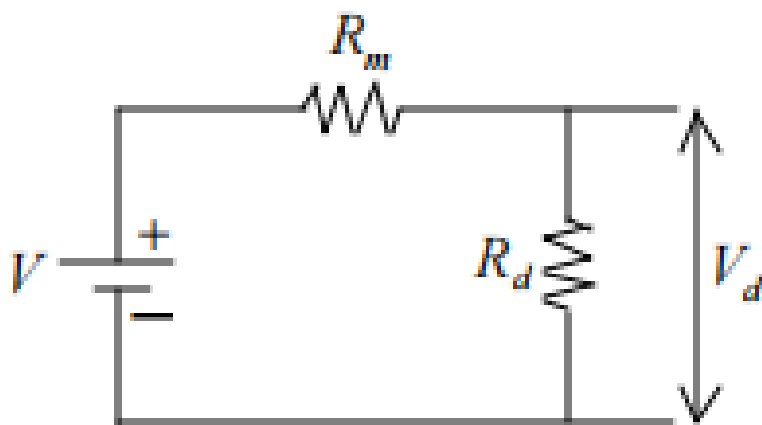


Figure 2: II-1-rasm. O'lja tomonidan hosil qilingan elektr quvvatini yirtqich tomonidan aniqlash modeli.



II-2-rasm.

Sezuvchi yirtqich, o'lja va atrofdagi dengiz suvini o'z ichiga olgan ekvivalent yopiq zanjir tizimi.

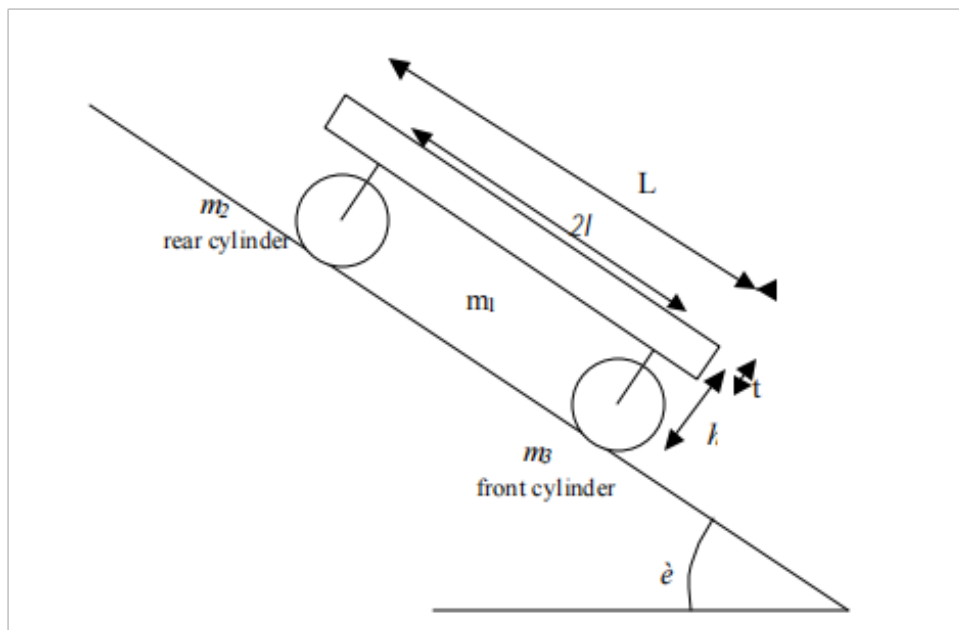
Rasmda V — o'lja tomonidan hosil qilingan elektr maydon ta'sirida detektor sharlari orasidagi kuchlanish farqi, R_m — atrofdagi dengiz suvining ichki qarshiligi (inner resistance). V_d va R_d esa mos ravishda detektor sharlari orasidagi kuchlanish farqi va yirtqich ichidagi detektor elementining qarshiligi.

Savollar:

1. Cheksiz muhitda nuqtaviy tok manbai I_s dan r masofada hosil bo'ladigan tok zichligi vektori $\vec{j}$ (birlik yuzaga to'g'ri keladigan tok) ni aniqlang. [1.5 ball]
2. $\vec{E} = \rho \vec{j}$ qonuniga asoslanib, o'lja tanasidagi ikki shar orasida oqayotgan I_s toki uchun detektor sharlarining o'rtasida (P nuqtasida) hosil bo'ladigan elektr maydon kuchi $\vec{E}_p$ ni aniqlang. [2.0 ball]

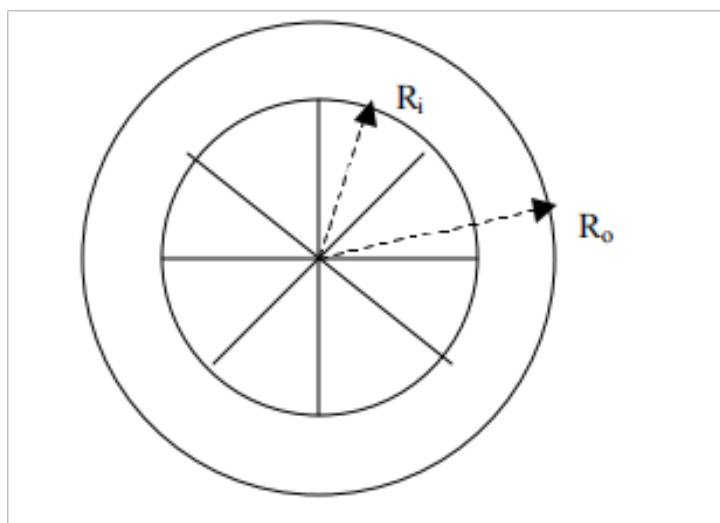
3. Xuddi shu I_s toki uchun o'ljadagi manba sharlari orasidagi kuchlanish farqi V_s ni [1.5 ball], ikki manba shari orasidagi qarshilik R_s ni [0.5 ball] va manba tomonidan ishlab chiqarilgan quvvat P_s ni [0.5 ball] aniqlang.
4. II-2-rasmdagi R_m [0.5 ball], V_d [1.0 ball] ni aniqlang va manbadan detektorga uzatiladigan quvvat P_d ni hisoblang [0.5 ball].
5. Detektorga uzatiladigan quvvat maksimal bo'lishi uchun R_d ning optimal qiymatini [1.5 ball] va shu maksimal quvvatni [0.5 ball] aniqlang.

III. Qiya yo'lda harakatlanayotgan og'ir transport vositasi (A Heavy Vehicle Moving on An Inclined Road)



III-1-rasm. Qiya yo'lda harakatlanayotgan og'ir transport vositasining soddalashtirilgan modeli.

Yuqoridagi rasmda qiya yo'lda (qiyalik burchagi ℓ) harakatlanayotgan og'ir transport vositasining (yo'l rolig'i) soddalashtirilgan modeli ko'rsatilgan. Transport vositasining ikkita g'ildiragi (biri orqada, biri oldinda) silindr shaklida. Har bir silindrning umumiy massasi M ($m_2 = m_3 = M$) bo'lib, ular tashqi radiusi R_0 , ichki radiusi $R_1 = 0.8R_0$ bo'lgan silindrik qobiq va umumiy massasi $0.2M$ bo'lgan sakkizta spitsadan iborat. Transport vositasi korpusini qo'llab-quvvatlovchi shassi (undercarriage) massasi e'tiborga olinmaydi. Silindrni III-2-rasmda ko'rsatilganidek modellashtirish mumkin. Transport vositasi tortishish va ishqalanish kuchlari ta'sirida qiya yo'ldan pastga qarab harakatlanmoqda. Oldingi va orqa silindrlar transport vositasiga nisbatan simmetrik joylashgan.



[H] III-2-rasm. Silindr modeli.

Silindr va yo'l orasidagi statik va kinetik ishqalanish koeffitsientlari mos ravishda μ_s va μ_k . Transport vositasi korpusining massasi $5M$, uzunligi L , qalinligi t . Oldingi va orqa

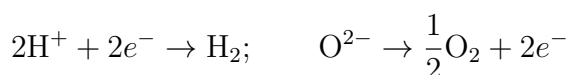
silindrlar orasidagi masofa $2l$, silindr markazidan korpus asosigacha bo'lgan masofa h . Silindr va uning o'qi orasidagi dumalab ishqalanish (rolling friction) e'tiborga olinmaydi.

Savollar:

1. Har bir silindrning inersiya momentini hisoblang. [1.5 ball]
2. Korpus, oldingi silindr va orqa silindrga ta'sir etuvchi barcha kuchlarni chizing. Ularning har biri uchun harakat tenglamalarini yozing. [2.5 ball]
3. Transport vositasi tinch holatdan boshlab, faqat tortishish ta'sirida erkin harakatlanadi deb faraz qilinadi. Tizimning barcha mumkin bo'lgan harakat turlarini ayting va ularning tezlanishlarini berilgan fizik kattaliklar orqali keltirib chiqaring. [4.0 ball]
4. Faraz qilaylik, transport vositasi tinch holatdan sof dumalab (pure rolling) d masofani bosib o'tgandan so'ng, yo'lning shunday qismiga kiradiki, bunda barcha ishqalanish koeffitsientlari μ_s va μ_k kichikroq o'zgarmas qiymatlarga tushadi va silindrlar sirpanishni boshlaydi. Transport vositasi jami s metr masofani bosib o'tgandan keyin har bir silindrning chiziqli va burchak tezliklarini hisoblang. Bu yerda d va s transport vositasining o'lchamlaridan ancha katta deb faraz qilinadi. [2.0 ball]

Nazariy asos (Background Theory)

Suvning elektrolizi quyidagi reaksiya bilan tavsiflanadi:



Reaksiya suvga botirilgan elektrodlar juftligiga elektr toki berilganda sodir bo'ladi. Reaksiyada hosil bo'lgan ikkala gaz ham ideal deb hisoblanadi.

Reaksiya natijasida hosil bo'lgan gazlardan biri ixtiyoriy shkalaga ega probirkada saqlanadi. O'tkazilgan umumiy zaryad va probirkadagi gaz hajmini bilgan holda e/k_B miqdori aniqlanishi mumkin, bu yerda e — elektron zaryadi, k_B — Boltsman doimiysi.

Yuqorida aytilgan maqsad uchun ushbu tajriba ikki qismga bo'lingan:

- **A qism:** Probirkadagi ixtiyoriy shkalani dinamik usul yordamida kalibrlash. Bu natija B qism uchun ishlatiladi.
- **B qism:** Suv elektrolizi orqali e/k_B fizik miqdorini aniqlash.

Ikkala tajribani (A qism va B qism) alifbo tartibida o'tkazish shart emas.

Quyidagi fizik miqdorlar ma'lum deb hisoblanadi:

- Erkin tushish tezlanishi, $g = (9.78 \pm 0.01) \text{ m/s}^2$
- Probirkaning ichki va tashqi diametrlari nisbati, $\alpha = 0.82 \pm 0.01$

Mahalliy temperatura T va bosim P qiymatlari tashkilotchi tomonidan beriladi.

Tajriba uchun asboblari va materiallar ro'yxati (A va B qismlari)

- Uch xil diametrli izolyatsiyalangan mis simlari:
 1. Kattaroq diametrli jigarrang
 2. Kichikroq diametrli jigarrang
 3. Ko'k
- Sozlanadigan kuchlanish manbai (0–60 V, maks. 1 A)
- Plastik idish va bir shisha suv
- Plastik qisqichli guruch blok (elektrodlarni sim izolyatsiyasiga zarar yetkazmasdan ushlab turish uchun)
- Raqamli sekundomer
- Multimetr (to'g'ri ishlatish tartibiga e'tibor bering)
- Probirkani tik ushlab turish uchun mo'ljallangan yog'och probirka ushlagichi
- Ko'p maqsadli pipetka
- Vertikal stand
- Belgilash uchun oq tuzatuvchi suyuqlik (white correction fluid) shishasi
- Kesgich (cutter)
- Qaychi Yelimli lenta (cellotape) rulosi
- Po'lat shar
- Elektrod sifatida ishlatiladigan zanglamaydigan po'lat plastinka bo'lagi
- Shkalali probirka
- Grafik qog'ozlari

Diqqat: Grafik qog'ozlari va tajriba asboblari (masalan, probirka) barcha shkalalar bir xil shkala birligiga ega, lekin millimetrda kalibrlanmagan.

Eksperiment (EXPERIMENT)

A qism: Probirkadagi ixtiyoriy shkalani kalibrlash

- Ixtiyoriy uzunlik shkalasini ma'lum shkalaga aylantirishga qodir bo'lgan dinamik usulni aniqlang.
- Tajribangizdagi o'lchanadigan miqdorlarni probirkaga bosilgan shkala orqali ifodalovchi ifodani yozing va tajriba qurilmasining eskizini chizing.
- Noma'lum uzunlik shkalasini aniqlash va kalibrlash uchun tajribangizdagi ma'lumotlarni to'plang va tahlil qiling.

B qism: e/k_B fizik miqdorini aniqlash

- Reaksiya davomida hosil bo'ladigan gazlardan birini ushlab qolish uchun probirkani to'g'ri joylashtirgan holda elektroliz tajribasini o'rnatish.
- Tajribada o'lchanadigan miqdorlar: vaqt t , tok I va suv sathining farqi Δh ni o'z ichiga olgan tenglamani keltirib chiqaring.
- Tajribangizdagi ma'lumotlarni to'plang va tahlil qiling. Oddiylik uchun, probirka ichidagi gaz bosimi butun tajriba davomida o'zgarmas deb faraz qilishingiz mumkin.
- e/k_B qiymatini aniqlang.

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JAVOB VARAQASI (ANSWER FORM)

A QISM

1. Tanlagan usulingizni ayting va ushbu usulning eksperimental qurilmasining eskizini chizing. [0.5 ball]
2. Tanlagan usulingizdagi o'lchanadigan miqdorlarni bog'laydigan ifodani yozing. [0.5 ball] Ushbu ifodani olishda ishlatilgan barcha yaqinlashishlarni ayting. [1.0 ball]
3. Ma'lumotlarni quyidagi tartibda to'plang va tashkil qiling: fizik miqdorlar, qiymatlar, birliklar. [1.0 ball]
4. Kalibrlash sifatini mustaqil o'lchangan ikki miqdor o'rtasidagi grafikni ko'rsatib, amal qilish oralig'ini belgilang. [0.5 ball]
5. Ixtiyoriy shkalaning eng kichik birligini millimetrlarda va o'lchashlardan kelib chiqadigan taxminiy xatolikni aniqlang. [1.5 ball]

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B QISM

1. Eksperimental qurilmaning eskizi. [1.0 ball]

2. Quyidagi ifodani keltirib chiqaring:

$$I\Delta t = \frac{e}{k_B} \frac{2P(\pi r^2)}{T} \Delta h \quad [1.5 \text{ ball}]$$

3. Ma'lumotlarni quyidagi formatda to'plang va tashkil qiling: fizik miqdorlar (qiymati, birliklari) [1.0 ball]

4. e/k_B qiymatini va uning taxminiy xatoligini aniqlang. [1.5 ball]

II. OPTIK QORIQUTI (OPTICAL BLACK BOX)

Tavsif (Description)

Ushbu masalada talabalar kub shaklidagi quti ichidagi noma'lum optik komponentlarni aniqlashlari kerak. Quti muhrlangan va faqat ikkita tor teshikka ega, ular qizil plastik qoplama bilan himoyalangan. Komponentlar tajribada kuzatilgan optik hodisalar yordamida aniqlanishi kerak. Plastik qoplama qatlamining kichik qalinlik effektini e'tiborsiz qoldiring.

Teshiklar markazidan o'tuvchi chiziqli qutining o'qi (axis) deb belgilanadi. Qizil plastik qoplamalardan tashqari, quyidagi ro'yxatdan uchta element (bir xil yoki har xil bo'lishi mumkin) mavjud:

- Ko'zgu (oyna) – tekis yoki sferik
- Linza – musbat yoki manfiy
- Yassi parallel yuzalarga ega shaffof plastinka (plane-parallel plate)
- Prizma
- Difraksion panjara

Shaffof komponentlar ishlatiladigan to'liq uzunligida sindirish ko'rsatkichi 1.47 bo'lgan materialdan tayyorlangan.

Mavjud asboblari (Apparatus available)

- To'liq uzunligi 670 nm bo'lgan lazer ko'rsatkich. **DIQQAT: LAZER NURIGA TO'G'RIDAN-TO'G'RI QARAMANG.**
- Optik rels (optical rail)
- Optik rels bo'ylab harakatlanuvchi kub uchun platforma
- Relsning uchiga biriktirilishi va boshqa o'lchovlar uchun undan ajratilishi mumkin bo'lgan ekran
- Grafik qog'ozlari varag'i (uni yopishqoq lenta bilan ekranga yopishtirish mumkin)
- Universal qisqich bilan jihozlangan vertikal stend va I-masalada ham ishlatilgan ixtiyoriy shakllari probirka

Diqqat: Grafik qog'ozlari va tajriba asboblariidagi barcha shakllar bir xil shkala birligiga ega, lekin millimetrda kalibrlanmagan.

Masala (The Problem)

Uchta komponentning har birini aniqlang va ularning tegishli spetsifikatsiyasini (xarakteristikasini) keltiring:

Har bir komponentning spetsifikatsiya parametrlari (masalan, fokus masofasi, egrilik radiusi) uchun yakuniy javoblaringizni millimetr, mikrometr yoki grafik qog'ozining shkalasida ifodalang.

Natijalarning aniqligini aniqlashingiz shart emas.

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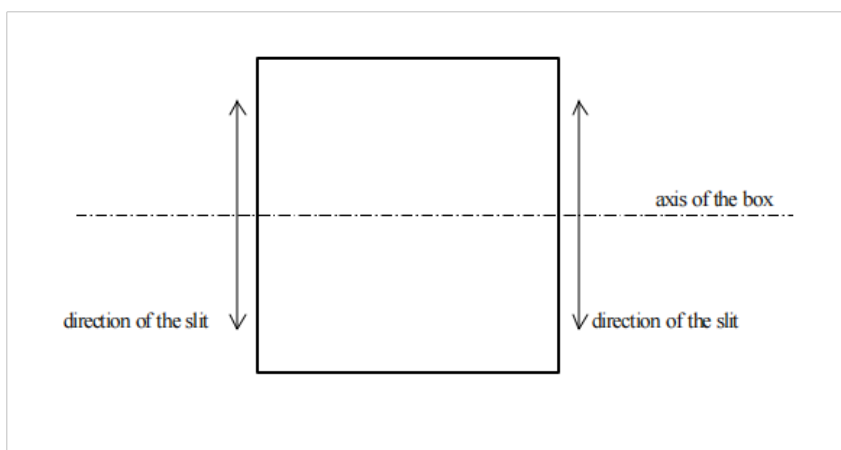
ANSWER FORM

JAVOB VARAQASI

1. Quti ichidagi optik komponentlarning turlarini yozing:

- (a) no.1. _____ [0.5 ball]
- (b) no.2. _____ [0.5 ball]
- (c) no.3. _____ [0.5 ball]

2. Quyidagi rasmda qutining ko'ndalang kesimi berilgan. Uchta komponent quti ichida qanday joylashganligini ko'rsatish uchun rasmga eskiz qo'shing. Eskizingizda har bir komponentni 1-javobdagi tartib raqami bilan belgilang.



- (a) 2-javobdagi optik komponentlarning joylashishiga oid qo'shimcha tafsilotlarni, masalan, burchak, komponentning teshikdan masofasi va komponentlarning orientatsiyasi yoki yo'nalishini ko'rsatuvchi qo'shimcha eskizlar bilan ta'minlang. [1.0 ball]

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- (a) 2-javobdagi optik komponentlarning joylashishiga oid qo'shimcha tafsilotlarni, masalan, burchak, komponentning teshikdan masofasi va komponentlarning orientatsiyasi yoki yo'nalishini ko'rsatuvchi qo'shimcha eskizlar bilan ta'minlang. [1.0 ball]

Qo'shimcha eskizlar shu sohaga chizilsin

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- (a) Kuzatilgan ma'lumotlarni jamlang [0.5 ball], tegishli formulani chizma yordamida keltirib chiqarib, 1-sonli optik komponentning spetsifikatsiyasini aniqlang [1.0 ball], so'ralgan spetsifikatsiyalarni hisoblang va quyidagi katakka javobingizni yozing [0.5 ball].

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- (a) Kuzatilgan ma'lumotlarni jamlang [0.5 ball], tegishli formulani chizma yordamida keltirib chiqarib, 2-sonli optik komponentning spetsifikatsiyasini aniqlang [1.0 ball], so'ralgan spetsifikatsiyalarni hisoblang va quyidagi katakka javobingizni yozing [0.5 ball].

Mumkin bo'lgan komponent turi	Talab qilinadigan spetsifikatsiya
Ko'zgu (mirror)	egrilik radiusi, ko'zgu o'qi va quti o'qi orasidagi burchak
Linza (lens)*	musbat yoki manfiy, fokus masofasi va quti ichidagi joylashuvi
Yassi parallel plastinka (plane-parallel plate)	qalinligi, plastinka va quti o'qi orasidagi burchak
Prizma	tepa burchagi (apex angle), uning og'diruvchi tomonlaridan biri va quti o'qi orasidagi burchak
Difrakzion panjara (diffraction grating)*	chiziqlar orasidagi masofa (line spacing), chiziqlarning yo'nalishi va quti ichidagi joylashuvi

* uning tekisligi quti o'qiga to'g'ri burchak ostida ekanligini anglatadi

1-komponent nomi	Spetsifikatsiya

2-komponent nomi	Spetsifikatsiya

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- (a) Kuzatilgan ma'lumotlarni jamlang [0.5 ball], tegishli formulani chizma yordamida keltirib chiqarib, 3-sonli optik komponentning spetsifikatsiyasini aniqlang [1.0 ball], so'ralgan spetsifikatsiyalarni hisoblang va quyidagi katakka javobingizni yozing [0.5 ball].

NAZARIY MUSOBAQA (THEORETICAL COMPETITION)

2002 yil 23-iyul, seshanba

I. Yerga kiruvchi radar (Ground-Penetrating Radar) – Yechim

1. Radar signalining materialdagi tezligi v_m

$$\omega t - \beta z = \text{const} \quad \Rightarrow \quad \beta z = -\text{const} + \omega t \quad (0.2 \text{ ball})$$

$$v_m = \frac{\omega}{\beta}$$

$$v_m = \frac{1}{\omega \left\{ \frac{\mu\epsilon}{2} \left[\left(1 + \frac{\sigma^2}{\epsilon^2 \omega^2} \right)^{1/2} + 1 \right] \right\}^{1/2}} \quad (0.4 \text{ ball})$$

$$v_m = \frac{1}{\left[\frac{\mu\epsilon}{2} (1 + 1) \right]^{1/2}} = \frac{1}{\sqrt{\mu\epsilon}} \quad (0.4 \text{ ball})$$

2. Yerdagi ob'ektni aniqlashning maksimal chuqurligi (teri qalinligi, skin depth) δ

So'nish konstantasiga teskari proporsional:

$$\delta = \frac{1}{\alpha} = \frac{1}{\omega \left\{ \frac{\mu\epsilon}{2} \left[\left(1 + \frac{\sigma^2}{\epsilon^2 \omega^2} \right)^{1/2} - 1 \right] \right\}^{1/2}}$$

$$\left(\frac{\sigma}{\omega\epsilon} \right)^2 \ll 1 \quad \text{shartida} \quad \left(1 + \frac{\sigma^2}{\epsilon^2 \omega^2} \right)^{1/2} \approx 1 + \frac{1}{2} \frac{\sigma^2}{\epsilon^2 \omega^2}$$

$$\delta = \frac{1}{\omega \left\{ \frac{\mu\epsilon}{2} \left[1 + \frac{1}{2} \frac{\sigma^2}{\epsilon^2 \omega^2} - 1 \right] \right\}^{1/2}} = \frac{1}{\omega \left\{ \frac{\mu\epsilon}{2} \cdot \frac{1}{2} \frac{\sigma^2}{\epsilon^2 \omega^2} \right\}^{1/2}} = \frac{1}{\omega \left\{ \frac{\mu\sigma^2}{4\epsilon\omega^2} \right\}^{1/2}}$$

$$\delta = \frac{1}{\omega \cdot \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \cdot \frac{1}{\omega}} = \frac{2}{\sigma} \sqrt{\frac{\epsilon}{\mu}}$$

Son qiymatlari:

$$\delta = \frac{2}{\sigma} \sqrt{\frac{\epsilon}{\mu}}, \quad \mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \quad \epsilon = \epsilon_r \epsilon_0, \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\sqrt{\frac{\epsilon}{\mu_0}} = \sqrt{\frac{\epsilon_0 \epsilon_r}{4\pi \times 10^{-7}}} = \sqrt{\frac{8.85 \times 10^{-12}}{4\pi \times 10^{-7}}} \sqrt{\epsilon_r} = \sqrt{7.04 \times 10^{-6}} \sqrt{\epsilon_r} \approx 2.65 \times 10^{-3} \sqrt{\epsilon_r}$$

$$\delta = \frac{2 \times 2.65 \times 10^{-3} \sqrt{\epsilon_r}}{\sigma} = \frac{5.31 \sqrt{\epsilon_r}}{\sigma} \text{ m}$$

bu yerda σ mS/m da. (0.5 ball)

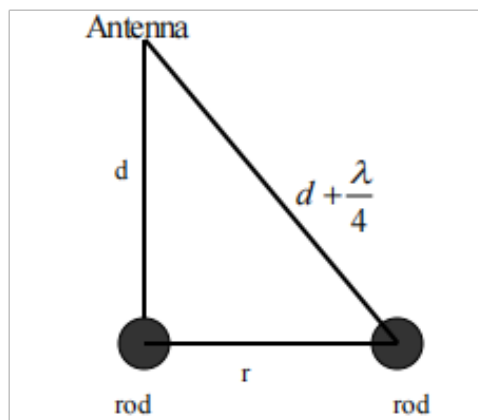
O'tkazuvchanlik $\sigma = 1.0$ mS/m va nisbiy dielektrik o'tkazuvchanlik $\epsilon_r = 9$ bo'lgan muhit uchun teri qalinligi:

$$\delta = \frac{5.31 \sqrt{9}}{1.0} = 5.31 \times 3 = 15.93 \text{ m} \quad (0.3 \text{ ball} + 0.2 \text{ ball})$$

NAZARIY MUSOBAQA (THEORETICAL COMPETITION)

2002 yil 23-iyul, seshanba

3. Lateral rezolyutsiya (yonma-yon ajratish)



*Antenna va ikki sterjen

$$r^2 + d^2 = \left(d + \frac{\lambda}{4}\right)^2$$

$$r = \left(\frac{\lambda d}{2} + \frac{\lambda^2}{16}\right)^{\frac{1}{2}} \quad (1.0 \text{ ball})$$

$r = 0.5$ m, $d = 4$ m:

$$\frac{1}{2} = \left(\frac{4\lambda}{2} + \frac{\lambda^2}{16}\right)^{\frac{1}{2}}, \quad \lambda^2 + 32\lambda - 4 = 0 \quad (0.5 \text{ ball})$$

To'liq uzunligi $\lambda = 0.125$ m. (0.3 ball + 0.2 ball)

Signalning muhitdagi tarqalish tezligi:

$$v_m = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}} = \frac{c}{\sqrt{\mu_r\epsilon_r}}, \quad c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

$$v_m = \frac{0.3}{\sqrt{\epsilon_r}} \text{ m/ns}, \quad \mu_r = 1 \text{ (magnit xususiyatga ega emas)} \quad (0.5 \text{ ball})$$

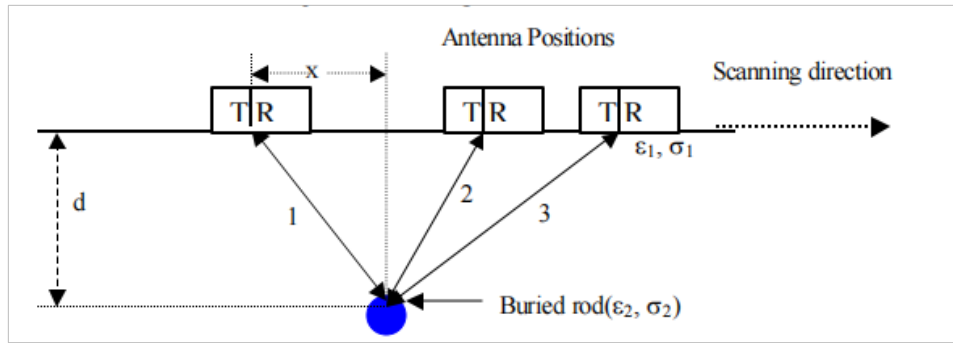
$$v_m = 0.1 \text{ m/ns} = 10^8 \text{ m/s}$$

Ikki sterjenni alohida ob'ekt sifatida ajratish uchun zarur minimal chastota:

$$f_{\min} = \frac{v}{\lambda} \quad (0.5 \text{ ball})$$

$$f_{\min} = \frac{0.3}{\sqrt{9}} \times 10^9 \text{ Hz} = 800 \text{ MHz} \quad (0.3 \text{ ball} + 0.2 \text{ ball})$$

4. Yer yuzasidagi turli nuqtalar uchun elektromagnit to'liqin yo'li



Ikki yo'l vaqti (travel time) t ning x ga bog'liqligi:

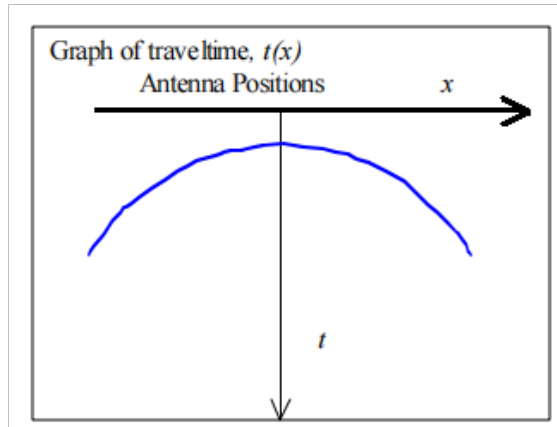
$$\left(\frac{t}{2}\right)^2 = \frac{d^2 + x^2}{v^2} \quad \text{yoki} \quad t(x) = \frac{2\sqrt{d^2 + x^2}}{v}$$

$$t(x) = \frac{2\sqrt{\epsilon_{1r}}}{0.3} \sqrt{d^2 + x^2}$$

$x = 0$ uchun:

$$100 \text{ ns} = 2 \times \frac{\sqrt{\epsilon_{1r}}}{0.3} d \quad (1.0 \text{ ball})$$

$\sqrt{\epsilon_{1r}} = \sqrt{9} = 3$ ekanligidan:



$$100 = 2 \times \frac{3}{0.3} d = 20d \quad \Rightarrow \quad d = 5 \text{ m} \quad (1.0 \text{ ball} + 0.5 \text{ ball})$$

II. Elektr signallarini sezish – Yechim

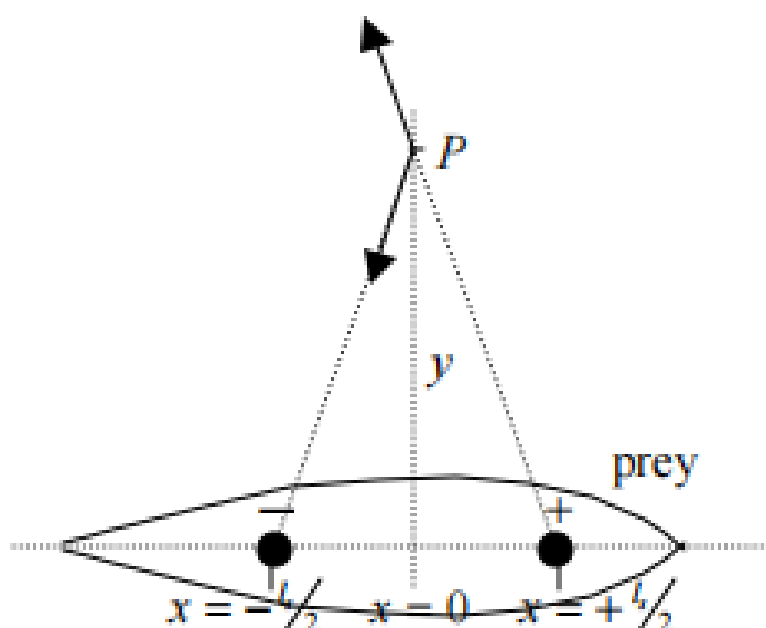
1.

Cheksiz izotrop muhitda nuqtaviy tok manbai I_s dan r masofadagi tok zichligi vektori

$$\vec{j} = \frac{I_s}{4\pi r^3} \vec{r}$$

[+1.5 ball] (vektor yozilmagan holda –0.5 ball)

2.



O'lja tanasi va atrofdagi dengiz suvining qarshiliklari bir xil deb faraz qilingan (o'ljani o'rab turgan chegara yo'qoladi), shuning uchun ikki shar qarshiligi ρ bo'lgan cheksiz izotrop muhitda joylashgandek bo'ladi. Kichik shar I_s tok chiqarayotgan bo'lsa, shar markazidan r masofadagi tok oqim zichligi ham

$$\vec{j} = \frac{I_s}{4\pi r^3} \vec{r}$$

Dengiz suvining qarshiligi ρ , shuning uchun r nuqtadagi maydon kuchi

$$\vec{E}(\vec{r}) = \rho \vec{j} = \frac{\rho I_s}{4\pi r^3} \vec{r} \quad [+0.2 \text{ ball}]$$

Modelda ikkita kichik shar mavjud. Biri ikkinchisiga nisbatan musbat kuchlanishda, shuning uchun I_s tok musbat zaryadlangan shardan manfiy zaryadlangan sharga oqadi. Ular bir-biridan l_s masofada joylashgan. $P(0, y)$ nuqtadagi maydon kuchi:

$$\vec{E}_p = \vec{E}_+ + \vec{E}_- \quad [+0.8 \text{ ball}]$$

$$= \frac{\rho I_s}{4\pi} \left[\frac{1}{\left(\left(\frac{l_s}{2}\right)^2 + y^2\right)^{3/2}} \left(-\frac{l_s}{2}\hat{i} + y\hat{j}\right) + \frac{1}{\left(\left(\frac{l_s}{2}\right)^2 + y^2\right)^{3/2}} \left(\frac{l_s}{2}\hat{i} - y\hat{j}\right) \right]$$

$$= \frac{\rho I_s}{4\pi} \frac{l_s(-\hat{i})}{\left(\left(\frac{l_s}{2}\right)^2 + y^2\right)^{3/2}}$$

$y \gg l_s$ bo'lganda yaqinlashish:

$$\vec{E}_p \approx \frac{\rho I_s l_s}{4\pi y^3} (-\hat{i}) \quad [+1.0 \text{ ball}]$$

3. Ikki shar orasidagi kuchlanish farqi V_s

Shar o'qlari bo'ylab (x o'qi) maydon kuchi:

$$\vec{E}(x) = \frac{\rho I_s}{4\pi} \left[\frac{1}{\left(x - \frac{l_s}{2}\right)^2} + \frac{1}{\left(x + \frac{l_s}{2}\right)^2} \right] (-\hat{i}) \quad [+0.5 \text{ ball}]$$

Kuchlanish farqi (manba sharlari orasidagi):

$$V_s = V_+ - V_- = - \int_{x=-l_s/2}^{x=+l_s/2} \vec{E}(x) \cdot d\vec{x}$$

Integrallash:

$$V_s = - \int_{-l_s/2}^{+l_s/2} \left\{ \frac{\rho I_s}{4\pi} \left[\frac{1}{\left(x - l_s/2\right)^2} + \frac{1}{\left(x + l_s/2\right)^2} \right] (-\hat{i}) \right\} \cdot (dx \hat{i})$$

$$= \frac{\rho I_s}{4\pi} \int_{-l_s/2}^{+l_s/2} \left[\frac{1}{\left(x - l_s/2\right)^2} + \frac{1}{\left(x + l_s/2\right)^2} \right] dx$$

Har bir hadni integrallaymiz:

$$\int \frac{dx}{\left(x - l_s/2\right)^2} = -\frac{1}{x - l_s/2}, \quad \int \frac{dx}{\left(x + l_s/2\right)^2} = -\frac{1}{x + l_s/2}$$

Chegaralar bo'yicha hisob:

$$\left[-\frac{1}{x - l_s/2} \right]_{-l_s/2}^{+l_s/2} = -\frac{1}{l_s/2 - l_s/2} + \frac{1}{-l_s/2 - l_s/2}$$

Birinchi had $\frac{1}{0}$ ga ajraladi (cheksiz). Bu integrallashni shar sirtlaridan tashqarida, to'g'ri usul bilan bajarish kerak. To'g'ri natija:

$$V_s \approx \frac{\rho I_s}{2\pi r_s} \quad [+0.5 \text{ ball}]$$

bu yerda r_s — sharlarning radiusi, $l_s \gg r_s$ shartida.

Ikki manba shari orasidagi qarshilik:

$$R_s = \frac{V_s}{I_s} = \frac{\rho}{2\pi r_s}$$

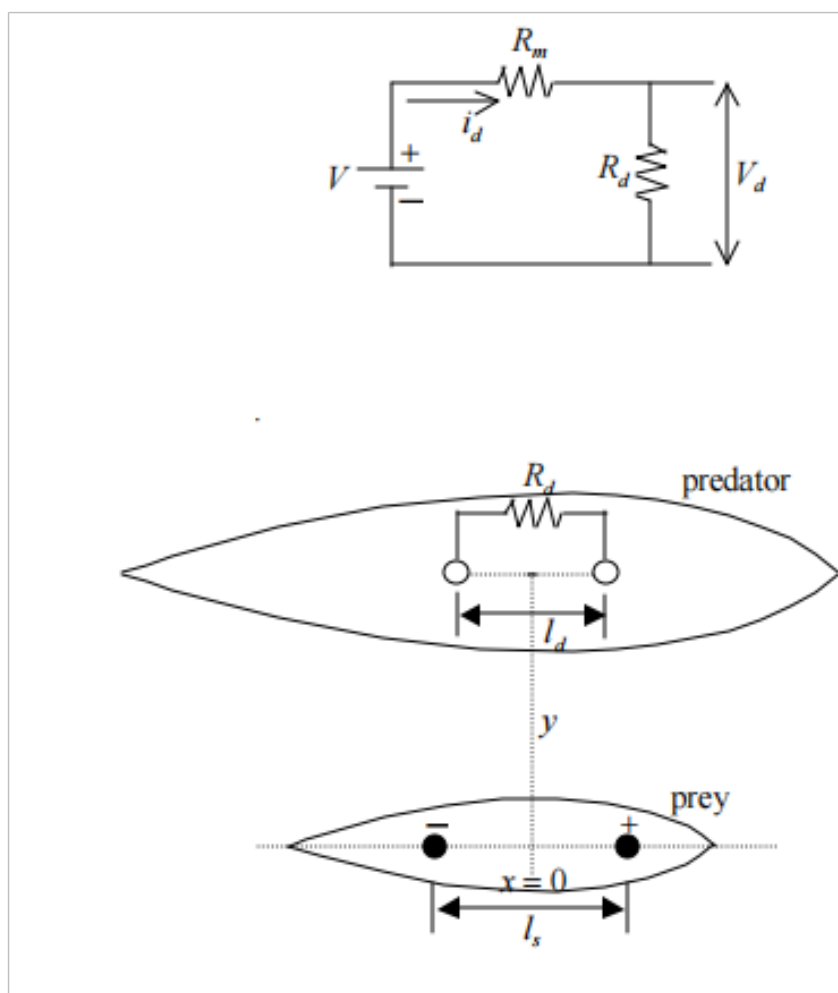
[+0.5 ball]

Manba tomonidan ishlab chiqarilgan quvvat:

$$P = I_s V_s = \frac{\rho I_s^2}{2\pi r_s}$$

[+0.5 ball]

4.



Ikki manba shari orasidagi qarshilikka o‘xshab, qarshiligi ρ bo‘lgan muhitning detektor sharlari orasidagi qarshiligi (har bir sharning radiusi r_d) quyidagicha:

$$R_m = \frac{\rho}{2\pi r_d}$$

[+0.5 ball]

l_d y dan ancha kichik bo'lgani sababli, detektor sharlari orasidagi elektr maydon kuchi o'zgarmas deb hisoblash mumkin, ya'ni:

$$E = \frac{\rho I_s l_s}{4\pi y^3} \quad [+0.2 \text{ ball}]$$

Shuning uchun muhitda detektor sharlari orasidagi kuchlanish farqi:

$$V = El_d = \frac{\rho I_s l_s l_d}{4\pi y^3} \quad [+0.3 \text{ ball}]$$

Detektor sharlari orasidagi kuchlanish farqi:

$$V_d = V \frac{R_d}{R_d + R_m} = \frac{\rho I_s l_s l_d}{4\pi y^3} \frac{R_d}{R_d + \frac{\rho}{2\pi r_d}}$$

[+0.5 ball]

Manbadan detektorga uzatiladigan quvvat:

$$P_d = i_d V_d = \frac{V}{R_d + R_m} V_d = \left(\frac{\rho I_s l_s l_d}{4\pi y^3} \right)^2 \left(\frac{R_d}{R_d + \frac{\rho}{2\pi r_d}} \right)^2$$

[+0.5 ball]

5.

P_d maksimal bo'lganda

$$R_t = \frac{R_d}{\left(R_d + \frac{\rho}{2\pi r_d} \right)^2} = \frac{R_d}{(R_d + R_m)^2} \text{ maksimal bo'ladi} \quad [+0.5 \text{ ball}]$$

Shuning uchun,

$$\frac{dR_t}{dR_d} = \frac{1(R_d + R_m)^2 - R_d \cdot 2(R_d + R_m)}{(R_d + R_m)^4} = 0 \quad [+0.5 \text{ ball}]$$

$$(R_d + R_m) - 2R_d = 0$$

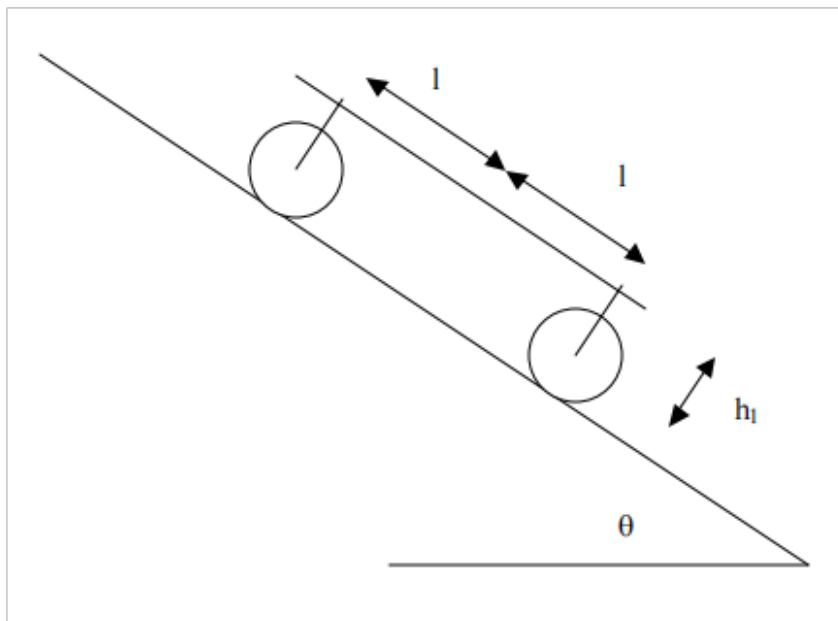
$$R_d^{\text{optimum}} = R_m = \frac{\rho}{2\pi r_d} \quad [+0.5 \text{ ball}]$$

Maksimal quvvat:

$$P_d^{\text{maksimum}} = \left(\frac{\rho I_s l_s l_d}{4\pi y^3} \right)^2 \frac{\pi r_d}{2\rho} = \frac{\rho(I_s l_s l_d)^2 r_d}{32\pi y^6}$$

[+0.5 ball]

III. Qiya yo'lda harakatlanayotgan og'ir transport vositasi – Yechim



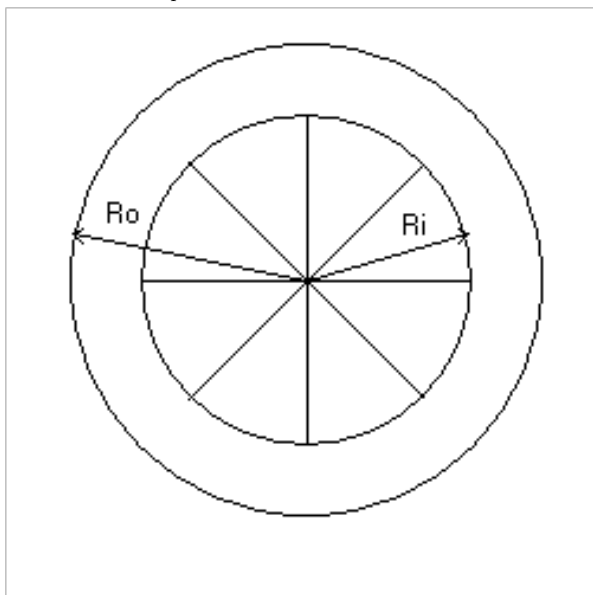
Modelni soddalashtirish uchun $h_l = h + 0.5t$, $R_o = R$

1. Silindrning inersiya momentini hisoblash

$$R_i = 0.8R_o$$

Silindr qismining massasi: $m_{\text{silindr}} = 0.8M$

Har bir spitsaning massasi: $m_{\text{spitsa}} = 0.025M$



$$\begin{aligned}
I &= \int r^2 dm = \int_{\text{silindr qobiq}} r^2 dm + \sum_{n=1}^8 \int_{\text{spitsa}} r^2 dm \\
\int_{\text{silindr qobiq}} r^2 dm &= 2\pi\sigma \int_{R_i}^{R_o} r^3 dr = 0.5\pi\sigma(R_o^4 - R_i^4) = 0.5m_{\text{silindr}}(R_o^2 + R_i^2) \\
&= 0.5(0.8M)R^2(1 + 0.64) = 0.656MR^2 \\
\int_{\text{spitsa}} r^2 dm &= \lambda \int_0^{R_i} r^2 dr = \frac{1}{3}\lambda R_i^3 = \frac{1}{3}m_{\text{spitsa}}R_i^2 = \frac{1}{3} \cdot 0.025M \cdot (0.64R^2) = 0.00533MR^2
\end{aligned}$$

Har bir g'ildirakning inersiya momenti:

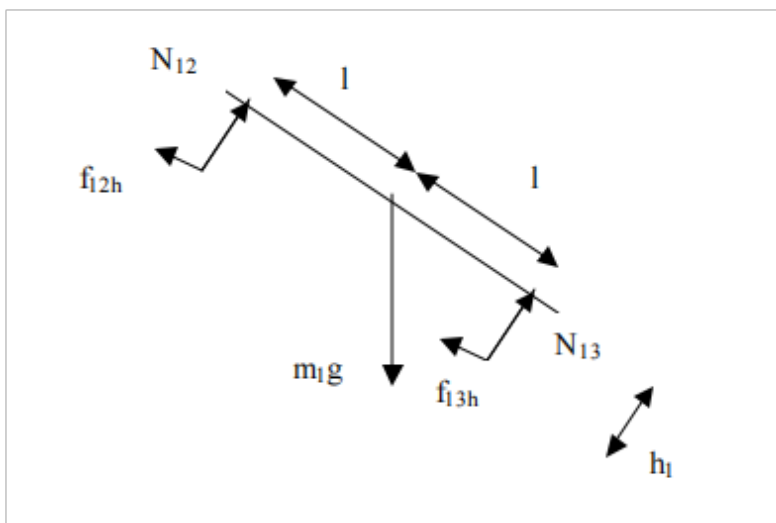
$$I = 0.656MR^2 + 8 \times 0.00533MR^2 = 0.7MR^2$$

(0.5 ball + 0.5 ball)

2. Kuchlar diagrammasi va muvozanat tenglamalari

Tahlilni soddalashtirish uchun tizimni uch qismga ajratamiz: ramka (1-qism, asosan tekis bir jinsli plastinka), orqa g'ildiraklar (ikkala g'ildirak birgalikda 2-qism), old g'ildiraklar (ikkala old g'ildirak birgalikda 3-qism).

1-qism: Ramka



Gorizontal o'q bo'yicha kuchlar muvozanati:

$$m_1g \sin \theta - f_{12h} - f_{13h} = m_1a$$

(0.2 ball)

Vertikal o'q bo'yicha kuchlar muvozanati:

$$m_1g \cos \theta = N_{12} + N_{13}$$

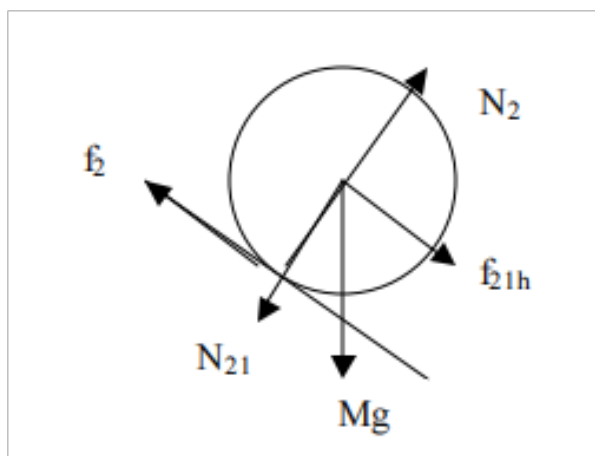
(0.2 ball)

O nuqtasiga nisbatan moment nol:

$$N_{12}l - N_{13}l + f_{12h}h_1 + f_{13h}h_1 = 0$$

(0.2 ball)

2-qism: Orqa g'ildirak



$$f_{21h} - f_2 + Mg \sin \theta = Ma$$

(0.15 ball)

$$N_2 - N_{21} - Mg \cos \theta = 0$$

(0.15 ball)

Sof dumalash (pure rolling) uchun:

$$f_2 R = I \alpha_2 = I \frac{a_2}{R} \quad \text{yoki} \quad f_2 = \frac{I}{R^2} a$$

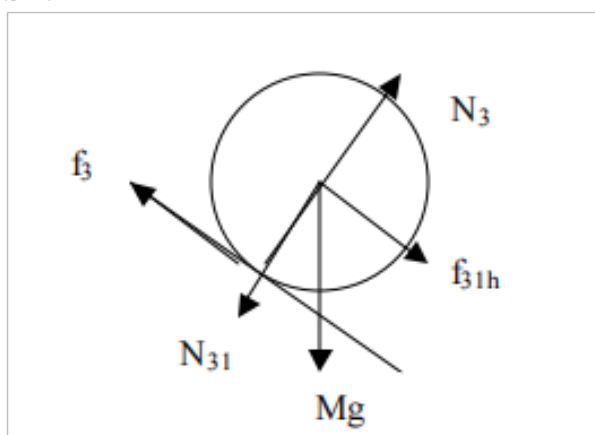
(0.2 ball)

Sirpanish bilan dumalash uchun:

$$f_2 = \mu_k N_2$$

(0.2 ball)

3-qism: Old g'ildirak



$$f_{31h} - f_3 + Mg \sin \theta = Ma$$

(0.15 ball)

$$N_3 - N_{31} - Mg \cos \theta = 0$$

Sof dumalash uchun: (0.15 ball)

$$f_3 = \frac{I}{R^2}a$$

Sirpanish bilan dumalash uchun: (0.2 ball)

$$f_3 = \mu_k N_3$$

(0.2 ball)

3.

(2), (5) va (9) tenglamalardan:

$$m_1 g \cos \theta = N_2 - m_2 g \cos \theta + N_3 - m_3 g \cos \theta$$

$$N_2 + N_3 = (m_1 + m_2 + m_3)g \cos \theta = 7Mg \cos \theta$$

(0.25 ball)

(3), (5) va (8) tenglamalardan:

$$(N_3 - Mg \cos \theta)l - (N_2 - Mg \cos \theta)l = h_1(f_2 + Ma - Mg \sin \theta + f_3 + Ma - Mg \sin \theta)$$

$$(N_3 - N_2) = \frac{h_1(f_2 + 2Ma - 2Mg \sin \theta + f_3)}{l}$$

(0.25 ball)

BARCHA G'ILDIRAKLAR SOF DUMALAYOTGAN HOL

(4) va (6) tenglamalardan:

$$f_{21h} - \frac{I}{R^2}a + Mg \sin \theta = Ma$$

(14) tenglamadan:

$$f_{21h} = \frac{I}{R^2}a + Ma - Mg \sin \theta \quad (0.2 \text{ ball})$$

(8) va (10) tenglamalardan:

$$f_{31h} = \frac{I}{R^2}a + Ma - Mg \sin \theta \quad (0.2 \text{ ball})$$

(1), (14) va (15) tenglamalardan:

$$5Mg \sin \theta - \left(\frac{I}{R^2}a + Ma - Mg \sin \theta \right) - \left(\frac{I}{R^2}a + Ma - Mg \sin \theta \right) = m_1 a$$

$$7Mg \sin \theta = \left(\frac{2I}{R^2} + 7M \right) a$$

$$a = \frac{7Mg \sin \theta}{7M + 2\frac{I}{R^2}} = \frac{7Mg \sin \theta}{7M + 2\frac{0.7MR^2}{R^2}} = 0.833 g \sin \theta \quad (0.35 \text{ ball})$$

$$\begin{aligned}
N_3 &= \frac{7M}{2}g \cos \theta + \frac{h_1}{l} \left[\left(M + \frac{I}{R^2} \right) \cdot 0.833g \sin \theta - Mg \sin \theta \right] \\
&= 3.5Mg \cos \theta + \frac{h_1}{l} [(M + 0.7M) \cdot 0.833g \sin \theta - Mg \sin \theta] \\
&= 3.5Mg \cos \theta + 0.41 \frac{h_1}{l} Mg \sin \theta \\
N_2 &= \frac{7M}{2}g \cos \theta - \frac{h_1}{l} \left[\left(\frac{I}{R^2} + M \right) \cdot 0.833g \sin \theta - Mg \sin \theta \right] \\
&= 3.5g \cos \theta - \frac{h_1}{l} \left[(0.7M + M) \cdot \frac{7Mg \sin \theta}{0.7M + 7M} - 2Mg \sin \theta \right] \\
&= 3.5g \cos \theta - 0.41 \frac{h_1}{l} Mg \sin \theta
\end{aligned}$$

Sof dumalash shartlari:

$$\begin{aligned}
f_2 &\leq \mu_s N_2 \quad \text{va} \quad f_3 \leq \mu_s N_3 \\
\frac{I_2}{R_2^2} a &\leq \mu_s N_2 \quad \text{va} \quad \frac{I_3}{R_3^2} a \leq \mu_s N_3
\end{aligned}$$

Chap tengsizlik:

$$\begin{aligned}
0.7M \times 0.833g \sin \theta &\leq \mu_s \left(3.5Mg \cos \theta - 0.41 \frac{h_1}{l} Mg \sin \theta \right) \\
\tan \theta &\leq \frac{3.5\mu_s}{0.5831 + 0.41\mu_s \frac{h_1}{l}}
\end{aligned}$$

O'ng tengsizlik:

$$\begin{aligned}
0.7M \times 0.833g \sin \theta &\leq \mu_s \left(3.5Mg \cos \theta + 0.41 \frac{h_1}{l} Mg \sin \theta \right) \\
\tan \theta &\leq \frac{3.5\mu_s}{0.5831 - 0.41\mu_s \frac{h_1}{l}} \quad (0.1 \text{ ball})
\end{aligned}$$

Barcha g'ildiraklar sirpanayotgan hol (All cylinders sliding)

(4) tenglamadan:

$$f_{21h} = Ma + \mu_k N_2 - Mg \sin \theta \quad (0.15 \text{ ball})$$

(8) tenglamadan:

$$f_{31h} = Ma + \mu_k N_3 - Mg \sin \theta \quad (0.15 \text{ ball})$$

(18) va (19) dan:

$$\begin{aligned}
5Mg \sin \theta - (Ma + \mu_k N_2 - Mg \sin \theta) - (Ma + \mu_k N_3 - Mg \sin \theta) &= m_1 a \\
a &= \frac{7Mg \sin \theta - \mu_k N_2 - \mu_k N_3}{7M} = g \sin \theta - \frac{\mu_k (N_2 + N_3)}{7M} \quad (0.2 \text{ ball})
\end{aligned}$$

$$N_2 + N_3 = 7Mg \cos \theta$$

Yuqoridagi ikki tenglamadan:

$$a = g \sin \theta - \mu_k g \cos \theta \quad (0.25 \text{ ball})$$

To'liq sirpanish shartlari sof dumalash shartlarining teskarisi:

$$f_2 = \mu_k N'_2, \quad f_3 = \mu_k N'_3$$

$$\frac{I_2}{R_2^2} a \geq \mu_k N'_2 \quad \text{va} \quad \frac{I_3}{R_3^2} a \geq \mu_k N'_3 \quad (0.2 \text{ ball})$$

bu yerda N'_2 va N'_3 barcha g'ildiraklar sof dumalayotgan holatda hisoblangan. (0.1 ball)
Natijada:

$$\tan \theta \geq \frac{3.5\mu_s}{0.5831 + 0.41\mu_s \frac{h_1}{l}} \quad \text{va} \quad \tan \theta \geq \frac{3.5\mu_s}{0.5831 - 0.41\mu_s \frac{h_1}{l}} \quad (0.2 \text{ ball})$$

Chap tengsizlik hal qiluvchi bo'ladi.

Bir g'ildirak sof dumalab, ikkinchisi sirpanayotgan hol (One cylinder pure rolling, another sliding)

Masalan, old g'ildiraklar (R_3) sof dumalaydi, orqa g'ildiraklar (R_2) sirpanaydi.

(4) tenglamadan:

$$f_{21h} = m_2 a + \mu_k N_2 - m_2 g \sin \theta \quad (0.15 \text{ ball})$$

(5) tenglamadan:

$$f_{31h} = m_3 a + \frac{I}{R^2} a - m_3 g \sin \theta \quad (0.15 \text{ ball})$$

(1), (22) va (23) dan:

$$m_1 g \sin \theta - \{m_2 a + \mu_k N_2 - m_2 g \sin \theta\} - \{m_3 a + \frac{I}{R^2} a - m_3 g \sin \theta\} = m_1 a$$

$$m_1 g \sin \theta + m_2 g \sin \theta + m_3 g \sin \theta - \mu_k N_2 = \left(\frac{I}{R^2} + m_3 \right) a + m_2 a + m_1 a$$

$$5Mg \sin \theta + Mg \sin \theta + Mg \sin \theta - \mu_k N_2 = (0.7M + M)a + Ma + 5Ma$$

$$a = \frac{7Mg \sin \theta - \mu_k N_2}{7.7M} = 0.9091g \sin \theta - \frac{\mu_k N_2}{7.7M} \quad (0.2 \text{ ball})$$

$$N_3 - N_2 = \frac{h_1}{l} \left(\mu_k N_2 + \frac{I}{R^2} a + 2Ma - 2Mg \sin \theta \right)$$

$$N_3 - N_2 = \frac{h_1}{l} \left(\mu_k N_2 + 2.7M \times 0.9091g \sin \theta - \frac{2.7\mu_k N_2}{7.7} - 2Mg \sin \theta \right)$$

$$N_3 - N_2 \left(1 + 0.65\mu_k \frac{h_1}{l} \right) = 0.4546Mg \sin \theta$$

$$N_3 + N_2 = 7Mg \cos \theta$$

Shuning uchun:

$$N_2 = \frac{7Mg \cos \theta - 0.4546Mg \sin \theta}{2 + 0.65\mu_k \frac{h_1}{l}} \quad (0.3 \text{ ball})$$

$$N_3 = 7Mg \cos \theta - \frac{7Mg \cos \theta - 0.4546Mg \sin \theta}{2 + 0.65\mu_k \frac{h_1}{l}}$$

So'ngra yuqoridagi natijalarni (24) tenglamaga qo'yib:

$$a = 0.9091g \sin \theta - \frac{\mu_k N_2}{7.7M} = 0.9091g \sin \theta - \frac{\mu_k}{7.7} \cdot \frac{7g \cos \theta - 0.4546g \sin \theta}{2 + 0.65\mu_k \frac{h_1}{l}} \quad (0.2 \text{ ball})$$

Qisman sirpanish shartlari (Conditions for partial sliding):

$$\begin{aligned} f_2 &\leq \mu_s N'_2 \quad \text{va} \quad f_3 \geq \mu_s N'_3 \\ \frac{I}{R^2} a &\leq \mu_s N'_2 \quad \text{va} \quad \frac{I}{R^2} a \geq \mu_s N'_3 \end{aligned}$$

$$(27) \quad \text{bu yerda } N'_2 \text{ va } N'_3 \text{ — sof dumalash holatidagi normal kuchlar.} \quad 0.25 \text{ ball}$$

4.

Faraz qilaylik, d metr dumalagandan so'ng barcha g'ildiraklar sirpanishni boshlaydi va qiya yo'lining oxirigacha (jami masofa s metr) shu holatda harakatlanadi. Faraz qilaylik, d masofa t_1 sekundda bosib o'tilgan.

$$v_{n1} = v_0 + a_1 t_1 = 0 + a_1 t_1 = a_1 t_1$$

$$d = v_0 t_1 + \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_1 t_1^2$$

$$t_1 = \sqrt{\frac{2d}{a_1}}$$

$$v_{n1} = a_1 \sqrt{\frac{2d}{a_1}} = \sqrt{2da_1} = \sqrt{2d \cdot 0.833g \sin \theta} = \sqrt{1.666 dg \sin \theta}$$

$$(28) \quad d \text{ metr dumalagandan keyingi burchak tezlik oldingi va orqa g'ildiraklar uchun bir xil:}$$

$$\omega_{n1} = \frac{v_{n1}}{R} = \frac{1}{R} \sqrt{1.666 dg \sin \theta}$$

$$(29) \quad 0.5 \text{ ball}$$

Keyin transport vositasi qiya yo'lining oxirigacha sirpanadi. t_2 sekund d nuqtadan yo'l oxirigacha harakatlanish vaqti bo'lsin.

$$v_{n2} = v_{n1} + a_2 t_2 = \sqrt{1.666 dg \sin \theta} + a_2 t_2$$

$$s - d = v_{n1} t_2 + \frac{1}{2} a_2 t_2^2$$

$$t_2 = \frac{-v_{n1} + \sqrt{v_{n1}^2 + 2a_2(s-d)}}{a_2}$$

$$v_{n2} = \sqrt{1.666 dg \sin \theta} - v_{n1} + \sqrt{v_{n1}^2 + 2a_2(s-d)}$$

(30)

0.4 ball

v_{n1} va a_2 ni oldingi natijalardan qo'yib yakuniy natijalarni olamiz.

Sirpanish paytida burchak tezlikka aylantiruvchi moment ta'sir qiladi:

4.

(31) tenglama:

$$\tau = \mu_k NR$$

$$\alpha = \frac{\tau}{I} = \frac{\mu_k NR}{I}$$

$$\omega_{12} = \omega_{11} + \alpha t_2 = \frac{1}{R} \sqrt{1.666 dg \sin \theta} + \frac{\mu_k NR - v_{11} + \sqrt{v_{11}^2 + 2a_2(s-d)}}{a_2}$$

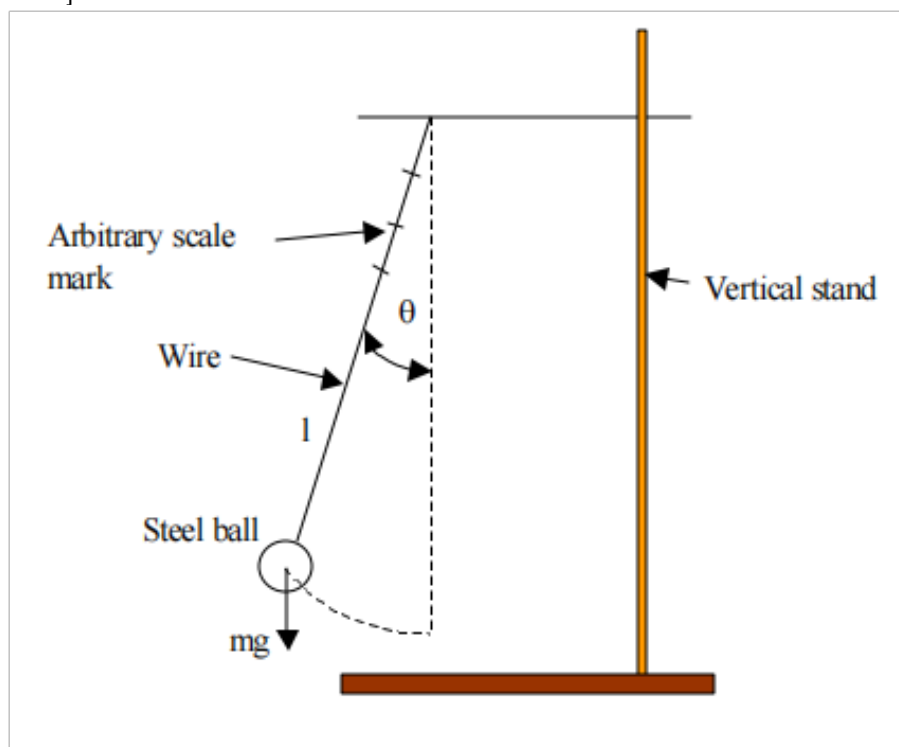
(0.6 ball)

Eksperimental musobaqa – Yechim (2002)

A QISM – Probirkadagi ixtiyoriy shkalani kalibrlash

1. [Jami 0.5 ball]

Ixtiyoriy shkalani kalibrlash uchun tanlangan eksperimental usul – oddiy mayatnik usuli.
[0.3 ball]



Eksperimental

qurilmaning eskizi.

- Ixtiyoriy shkala belgisi
- l
- Po'lat shar
- mg
- θ
- Vertikal stend

[0.2 ball]

2. [Jami 1.5 ball]

O'lchanadigan miqdorlarni bog'laydigan ifoda:

[0.5 ball]

$$T_{\text{osc}} = 2\pi\sqrt{\frac{l}{g}}; \quad T_{\text{osc}}^2 = 4\pi^2\frac{l}{g}$$

Yaqinlashishlar (approximations):

- $\sin \theta \approx \theta$

[0.5 ball]

- Matematik mayatnik (ip massasi « shar massasi, shar radiusi « ip uzunligi) [0.5 ball]
- Ipning egiluvchanligi, havo ishqalanishi va boshqalar [0.1 ball, faqat yuqoridagi ikkita asosiy nuqta berilmagan bo'lsa]

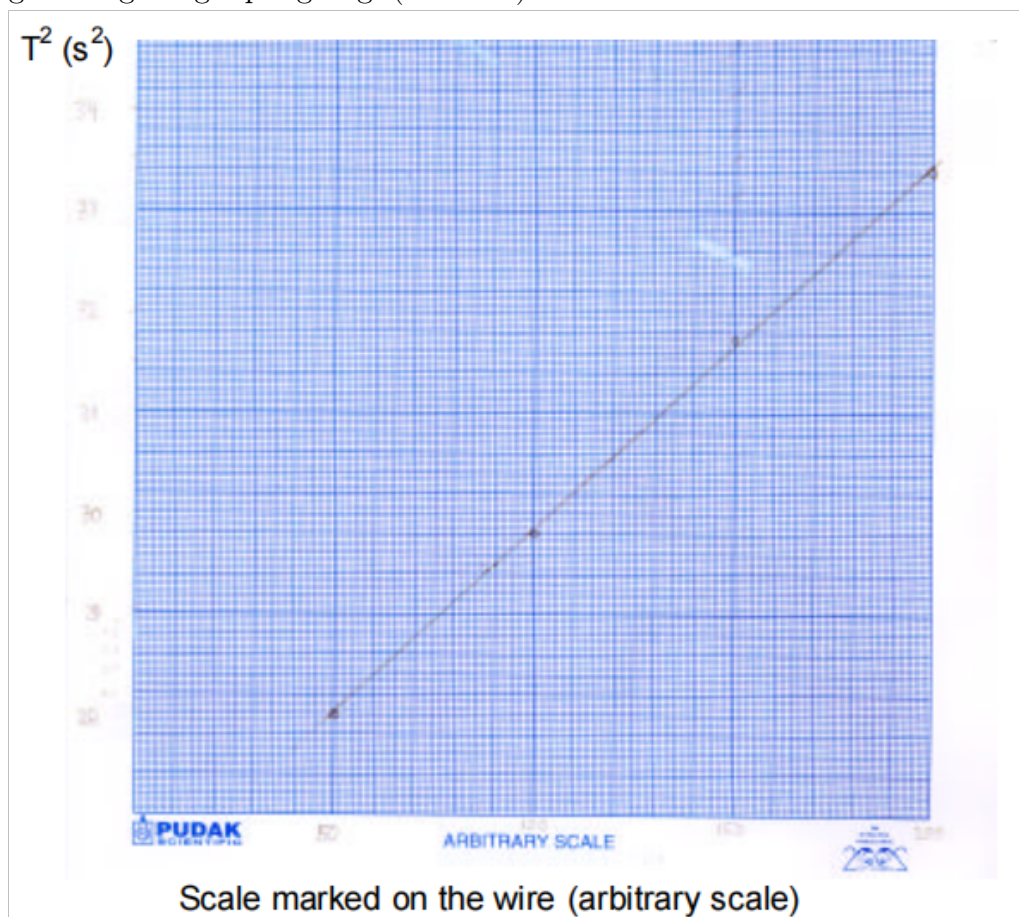
3. [Jami 1.0 ball]

Oddiy mayatnik tajribasidan ma'lumotlar namunasi:

- Sikl soni ≥ 20 [0.2 ball]
- T farqi ≥ 0.01 s [0.4 ball]
- Ma'lumotlar soni ≥ 4 [0.4 ball]

4. [Jami 0.5 ball]

T^2 ning shkalaga bog'liqlik grafigi (chiziladi).



5. Ixtiyoriy shkalaning eng kichik birligini millimetrlarda aniqlash [Jami 1.5 ball]

$$T_{\text{osc1}}^2 = \frac{4\pi^2}{g} L_1, \quad T_{\text{osc2}}^2 = \frac{4\pi^2}{g} L_2$$

$$T_{\text{osc1}}^2 - T_{\text{osc2}}^2 = \frac{4\pi^2}{g} (L_1 - L_2) = \frac{4\pi^2}{g} \Delta L$$

$$\Delta L = \frac{g}{4\pi^2} (T_{\text{osc1}}^2 - T_{\text{osc2}}^2) \quad \text{yoki boshqa ekvivalent ifoda}$$

Ixtiyoriy shkalaning eng kichik birligining o'rtacha qiymati $\bar{I} = 0.89 \text{ mm}$ [0.5 ball]
 O'lchashdan kelib chiqadigan taxminiy xatolik: [0.5 ball]
 Standart og'ish (standard deviation): [0.5 ball]

$$\Delta I = \sqrt{\frac{\sum_{i=1}^6 (I - \bar{I})^2}{N - 1}} = \sqrt{\frac{0.003}{5}} = 0.02 \text{ mm}$$

(boshqa asosli usullar ham qo'llanilishi mumkin)

3-komponent nomi	Spetsifikatsiya

Table 1: Mayatnik ma'lumotlari

50 sikl uchun t (s)	Davr T (s)	Ipga belgilangan shkala (ixtiyoriy birlik)
91.47	1.83	200
89.09	1.78	150
86.45	1.73	100
83.8	1.68	50

Table 2: T^2 va shkala

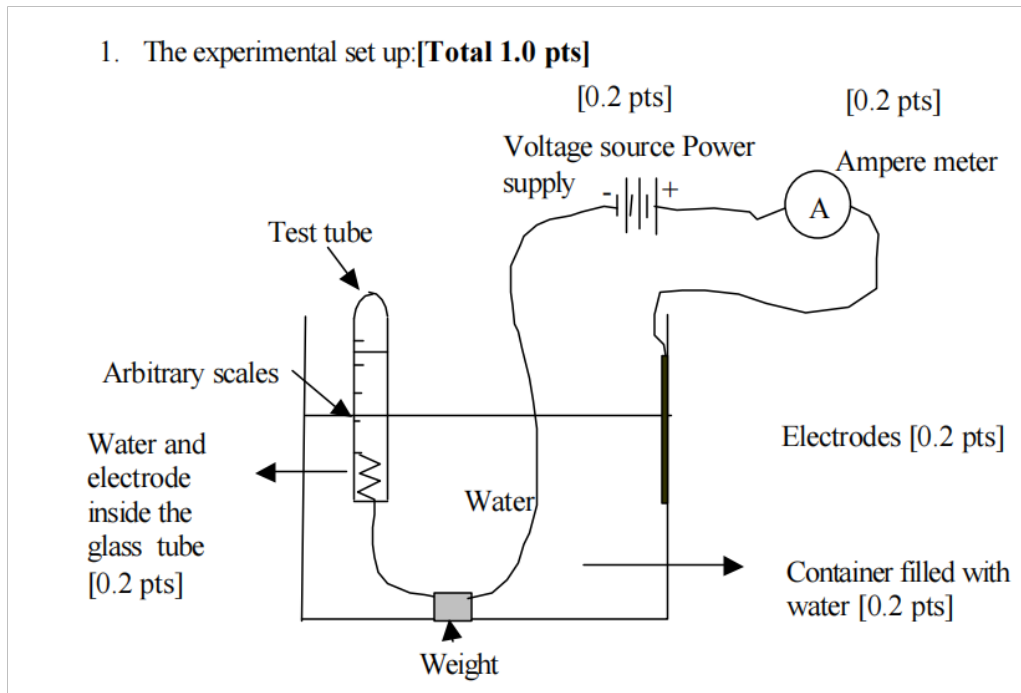
Davr T (s)	Shkala (ixtiyoriy birlik)	T^2 (s ²)
1.83	200	3.35
1.78	150	3.17
1.73	100	2.99
1.68	50	2.81

No.	Hisoblangan ΔL (m)	ΔL ixtiyoriy shkalada	Eng kichik birlik qiymati (mm)
1. 0.85	$T_1^2 - T_2^2 = 0.171893 \text{ s}^2$	0.042626	50
2. 0.89	$T_1^2 - T_3^2 = 0.357263 \text{ s}^2$	0.088595	100
3. 0.89	$T_1^2 - T_4^2 = 0.537728 \text{ s}^2$	0.133347	150
4. 0.92	$T_2^2 - T_3^2 = 0.18537 \text{ s}^2$	0.045968	50
5. 0.91	$T_2^2 - T_4^2 = 0.365835 \text{ s}^2$	0.09072	100
6. 0.90	$T_3^2 - T_4^2 = 0.180465 \text{ s}^2$	0.044752	50

No.	Eng kichik birlik qiymati (mm)	$I - \bar{I}$	$(I - \bar{I})^2$
1.	0.85	-0.04	0.0016
2.	0.89	0	0
3.	0.89	0	0
4.	0.92	0.03	0.0009
5.	0.91	0.02	0.0004
6.	0.90	0.01	0.0001

B QISM – e/k_B ni aniqlash

1. Eksperimental qurilmaning eskizi [Jami 1.0 ball]



Elektroliz qurilmasi.

- Kuchlanish manbai (Voltage source / Power supply) [0.2 ball]
- Ampermetr [0.2 ball]
- Probirka (Test tube) [0.2 ball]
- Suv va elektrodlar [0.2 ball]
- Suv to'ldirilgan idish [0.2 ball]

2. t , I va Δh ni bog'lovchi tenglamani keltirib chiqarish [Jami 1.5 ball]

$$I = \frac{\Delta Q}{\Delta t}$$

Reaksiyadan: $2H^+ + 2e^- \rightarrow H_2$. ΔN molekula hosil bo'lishi uchun zarur zaryad:
 $\Delta Q = 2e\Delta N$ [0.2 ball]

$$I = \frac{\Delta N \cdot 2e}{\Delta t} [0.5ball]$$

Ideal gaz tenglamasi: $P\Delta V = \Delta N k_B T$ [0.5 ball]

$$P\Delta V = \frac{I\Delta t}{2e} k_B T$$

$\Delta V = \Delta h \cdot \pi r^2$ ekanligidan:

$$P \cdot \Delta h(\pi r^2) = \frac{I\Delta t}{2} \cdot \frac{k_B}{e} T [0.2ball]$$

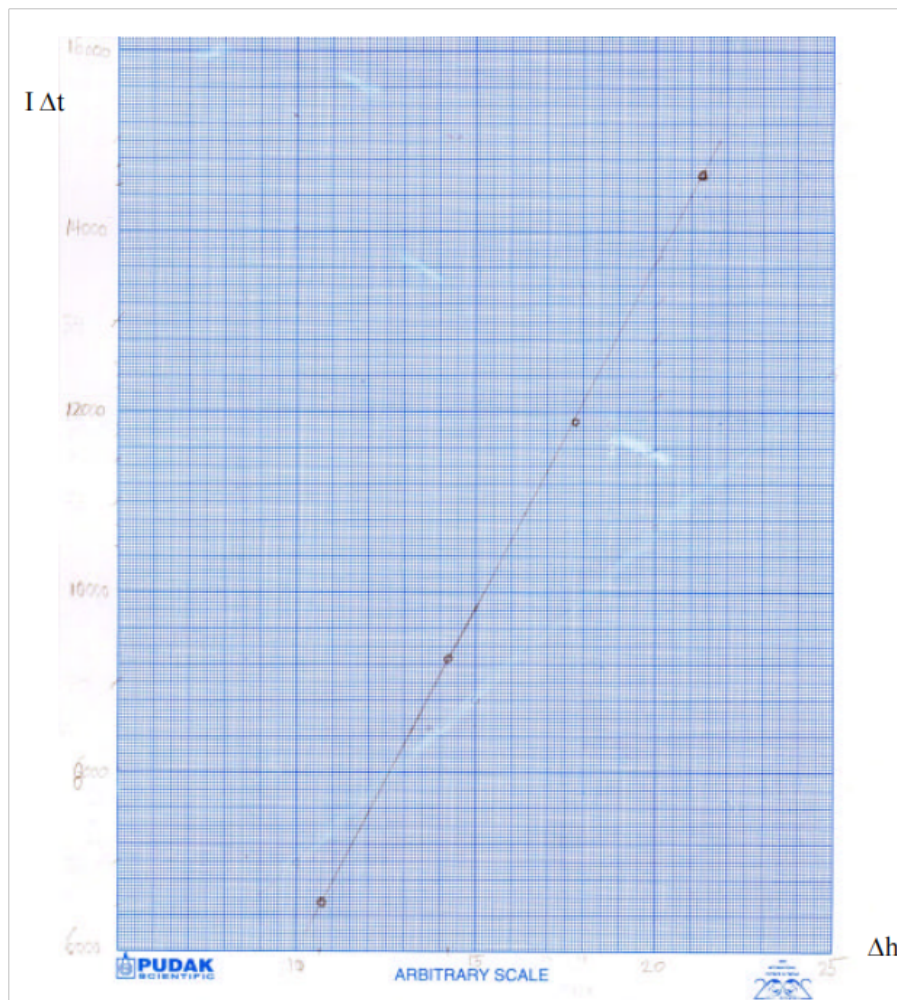
$$I\Delta t = \frac{e}{k_B} \cdot \frac{2P(\pi r^2)}{T} \Delta h [0.1ball]$$

3. Eksperimental ma'lumotlar [Jami 1.0 ball]

- Probiraning aylanasi $\phi = 46$ ixtiyoriy shkala [0.3 ball]
- Δh uchun tanlangan qiymatlar (≥ 4 shkala birligi) – suv sathini o'qishdagi xatolikni kamaytirish uchun; $I \leq 4$ mA – xalaqitni kamaytirish uchun [0.3 ball]
- Ma'lumotlar soni ≥ 4 [0.4 ball]

Atrof muhit sharoiti: $T = 300$ K, $P = 1.00 \times 10^5$ Pa.

4. e/k_B qiymatini aniqlash [Jami 1.5 ball]



(Keyingi hisoblashlar – e/k_B ning o'rtacha qiymati va xatoligi – ma'lumotlar asosida bajariladi. Natija: $\frac{e}{k_B} = \dots$)

Grafikdan olingan qiyalik (slope): 763.94

$$\frac{e}{k_B} = \frac{763.94 \times 300 \times \pi}{2 \times 10^5 \times (23 \times 0.89 \times 10^{-3} \times 0.82)^2} = 1.28 \times 10^4 \text{ Coulomb K/J}$$

[1.0 ball]

Shu natijaga alternativa (bir xil ball):

e/k_B ning o'rtacha qiymati: 1.07×10^4 Coulomb K/J

[1.0 ball]

Taxminiy xatolik: olingan standart chetlanish 0.66×10^3 Coulomb K/J.

[0.5 ball]

(Boshqa asosli xatolik olchovlari ham qollanilishi mumkin)

Table 3: Ma'lumotlar			
No.	Δh (ixtiyoriy shkala)	I (mA)	Δt (s)
1	12	4.00	1560.41
2	16	4.00	2280.61
3	20	4.00	2940.00
4	24	4.00	3600.13

Table 4: Hisoblash jadvali					
No.	Δh (shkala)	Δh (mm)	I (mA)	Δt (s)	$I\Delta t$ (C)
1	12	10.68	4.00	1560.41	6241.64
2	16	14.24	4.00	2280.61	9120.48
3	20	17.80	4.00	2940.00	11760.00
4	24	21.36	4.00	3600.13	14400.52

Table 5: $I\Delta t$ ning Δh ga bog'liqligi (yuqoridagi jadval asosida)

Δh (mm)	$I\Delta t$ (C)
10.68	6241.64
14.24	9120.48
17.80	11760.00
21.36	14400.52

No.	Δh (mm)	$I\Delta t$ (C)	Qiyalik (slope)	e/k_B
1	10.68	6241.64	584.4232	9774.74
2	14.24	9120.48	640.4831	10712.37
3	17.80	11760.00	660.6742	11050.07
4	21.36	14400.52	674.1816	11275.99

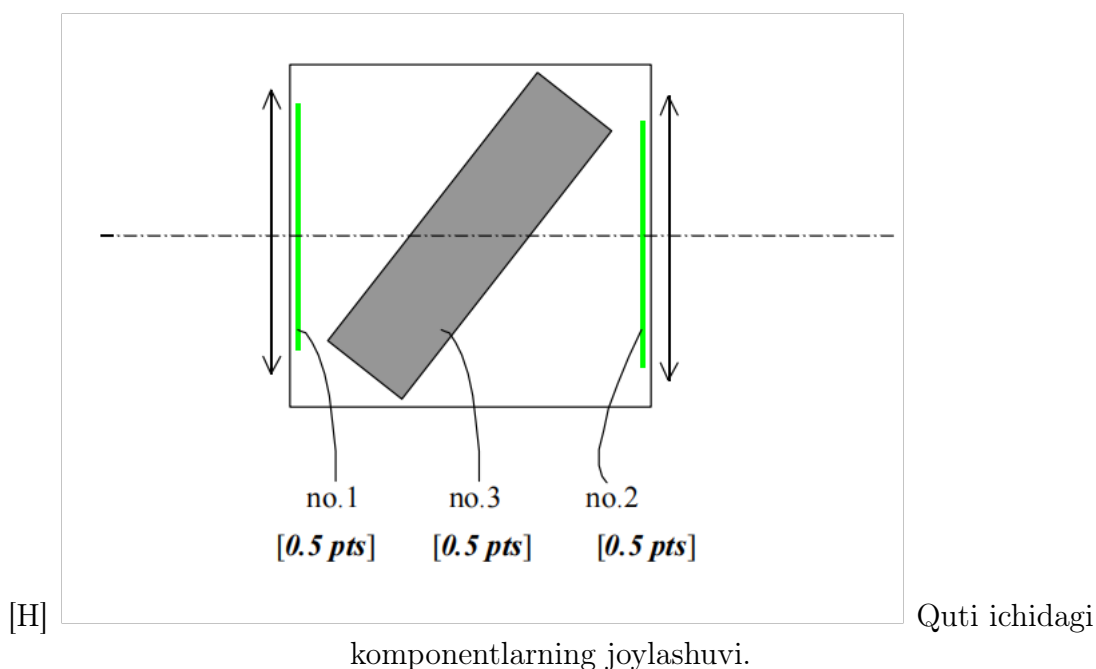
No.	e/k_B	Farq	Farq kvadrati
1	9774.74	-928.55	862205.5
2	10712.37	9.08	82.39
3	11050.07	346.78	120256.9
4	11275.99	572.70	327984.9

Eksperimental masala 2 – yechim (Optik qora quti)

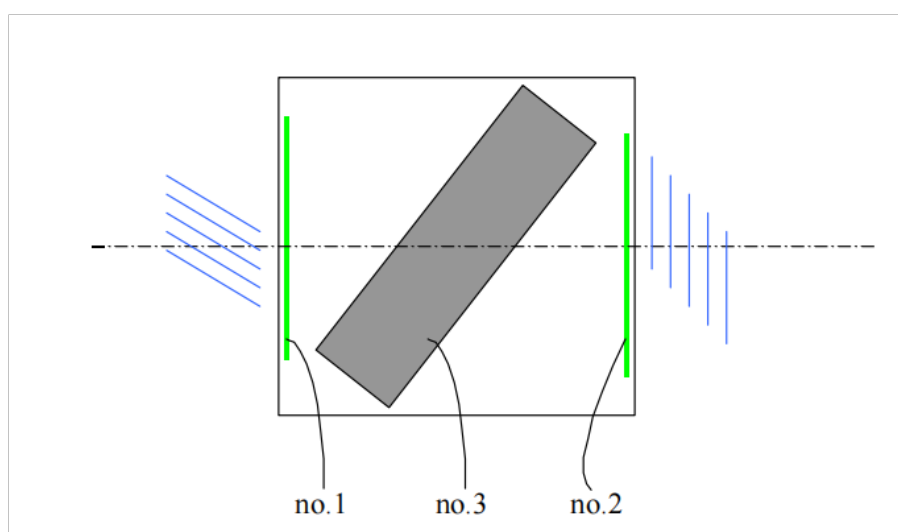
1. Optik komponentlar [jami 1.5 ball]

- No.1 – Difraksion panjara (diffraction grating) [0.5 ball]
- No.2 – Difraksion panjara [0.5 ball]
- No.3 – Yassi parallel plastinka (plane-parallel plate) [0.5 ball]

2. Qutining koʻndalang kesimi [jami 1.5 ball]



3. Qoʻshimcha maʼlumotlar [jami 1.0 ball]



- 1-panjaradan chap devorgacha masofa amalda nol. [0.2 ball]
- 1-panjaradagi chiziqlar yoriqqa (slit) toʻgʻri burchak ostida. [0.3 ball]
- 2-panjaradan ong devorgacha masofa amalda nol. [0.2 ball]

- 2-panjaradagi chiziqlar yoriqqa parallel. [0.3 ball]

4. Difraksion panjara (No.1) [jami 2.0 ball]

Chizma va belgilar toliq bolishi kerak. [0.6 ball]

Yol farqi (path length difference):

$$\Delta = d \sin \theta, \quad d = \text{panjara chiziqlari orasidagi masofa}$$

Difraksion tartib:

$$\Delta = m\lambda$$

Shuning uchun birinchi tartib ($m = 1$) uchun:

$$\sin \theta = \frac{\lambda}{d} [0.4ball]$$

Kuzatuv ma'lumotlari (kamida 3 nuqta): [0.5 ball]

$\tan \theta$	θ	$\sin \theta$
0.34	18.78^0	0.3219
0.32	17.74^0	0.3048
0.32	17.74^0	0.3048

(Izoh: panjara chiziqlar oraligining haqiqiy qiymati $2.0 \mu\text{m}$, natijaning ogishi $\leq 10\%$.)
[0.4 ball + 0.1 ball]

5. Difraksion panjara (No.2) [jami 2.0 ball]

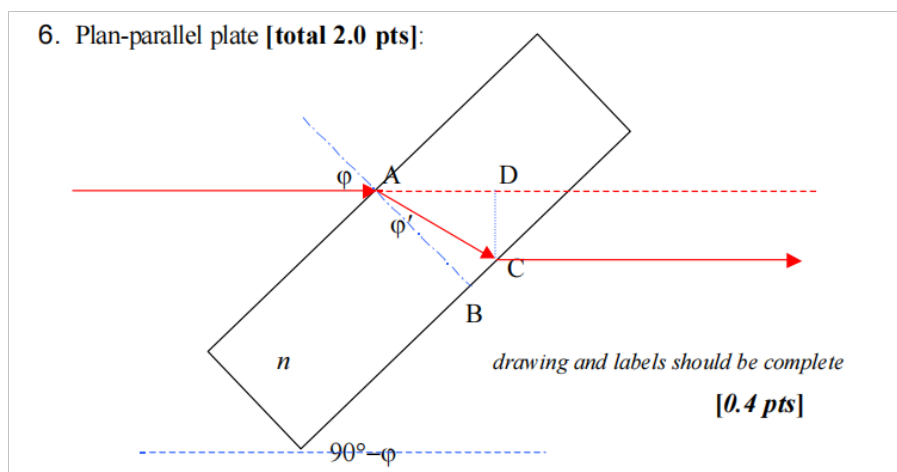
Formulani keltirib chiqarish yuqoridagi 4-bandga o'xshash. [1.0 ball]

Kuzatuv ma'lumotlari (kamida 3 nuqta): [0.5 ball]

$\tan \theta$	θ	$\sin \theta$
1.04	46.12^0	0.7208
0.96	43.83^0	0.6925
1.08	47.20^0	0.7330

(Izoh: haqiqiy qiymat $1.0 \mu\text{m}$, ogish $\leq 10\%$.) [0.4 ball + 0.1 ball]

6. Yassi parallel plastinka (No.3) [jami 2.0 ball]



Snellius qonuni:

$$\sin \varphi = n \sin \varphi', \quad \varphi' = \angle BAC$$

Plastinka ichidagi yol uzunligi:

$$AC = \frac{AB}{\cos \varphi'}, \quad AB = h = \text{plastinka qalinligi}$$

Nurning siljishi (beam displacement):

$$CD = t = AC \cdot \sin \angle CAD, \quad \angle CAD = \varphi - \varphi'$$

Shuning uchun:

$$t = h \sin \varphi \left[1 - \frac{\cos \varphi}{\sqrt{n^2 - \sin^2 \varphi}} \right] [0.6 \text{ ball}]$$

Kuzatuv ma'lumotlari:

[0.5 ball]

φ	t (ixtiyoriy shkala)
0	0
49°	7.3

(Nur va quti oqi orasidagi burchak 49°)

(Izohlar: haqiqiy qalinlik 20 mm, haqiqiy burchak 52°, natijaning ogishi $\leq 20\%$.) [0.2 ball + 0.3 ball]

1-komponent nomi	Spetsifikatsiya
Difraksion panjara	Chiziqlar orasidagi masofa = 2.16 μm Chiziqlar yoriqqa togri burchak ostida

2-komponent nomi	Spetsifikatsiya
Difraksion panjara	Chiziqlar orasidagi masofa = 0.936 μm Chiziqlar yoriqqa parallel

3-komponent nomi	Spetsifikatsiya
Yassi parallel plastinka	Qalinligi = 17.9 mm Quti oqiga nisbatan burchagi = 49°